

Lab: Intrusion Detection Systems

Estimated time: 30 minutes

Objective

The objective of this lab is to provide a comprehensive understanding of network intrusion detection by simulating and analyzing common types of network attacks on a Windows-based system. Specifically, you will execute a Ping of Death attack and a port scanning attack to observe how these activities impact a network. Additionally, you'll explore how a signature-based Intrusion Detection System (IDS) detects these attacks.

You will develop a deeper understanding of the techniques attackers use to disrupt network operations and the importance of effective intrusion detection systems in identifying and mitigating such threats in real time. This lab reinforces the necessity and value of maintaining by recognizing and responding to various forms of malicious activity to maintain a secure network environment.

Prerequisites

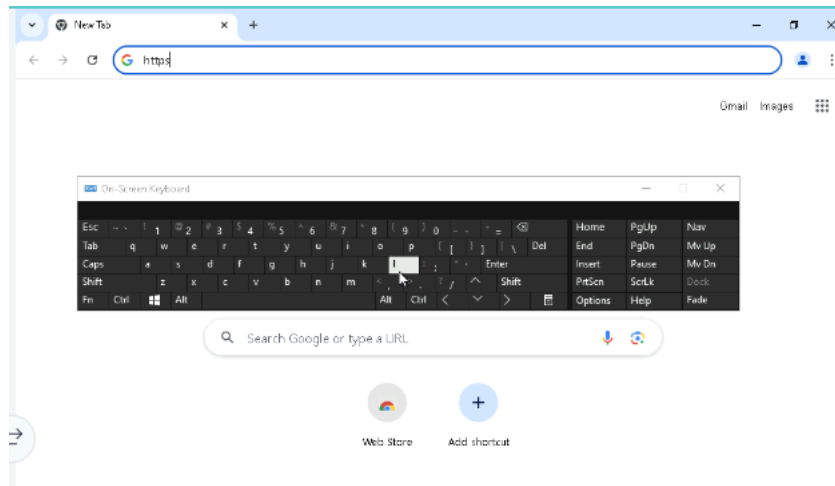
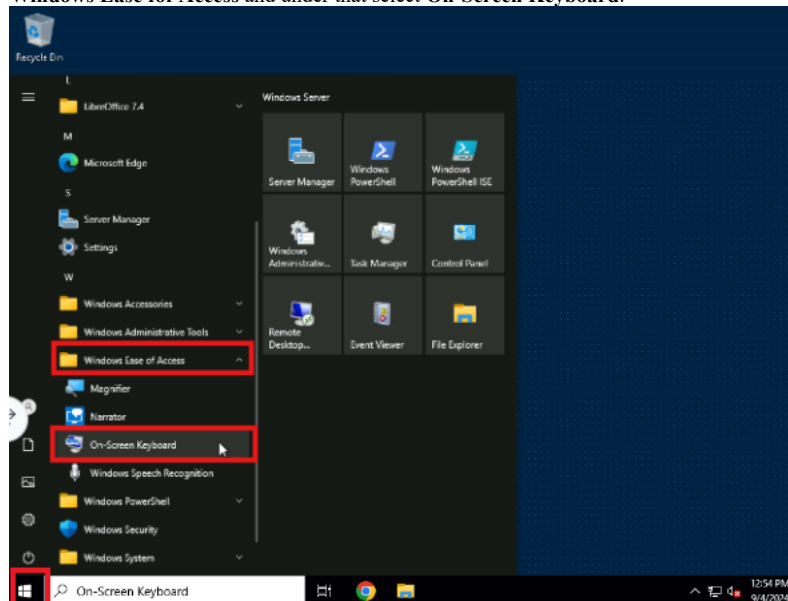
To successfully complete this lab you'll need the following skills and software:

- Basic understanding of networking concepts (ICMP, TCP/IP)
- Python installed on your Windows machine
- Nmap installed on your Windows Machine
- Port Scan Attacker and Detector is available on Machine

These lab instructions are written for a virtual Windows environment. If you choose to try this lab on your own machine, some screens might look slightly different as you work through the lab steps.

If you wish to view the instructions in the virtual windows environment you can access it by typing the following link into your browser: <https://bit.ly/3ThYIBM>

Note: If you are unable to type in the virtual windows environment you can use the **On-Screen Keyboard** from the environment. Click on the windows icon and look for **Windows Ease for Access** and under that select **On-Screen Keyboard**.



Important! When using the virtual lab environment you must either perform a web search for the application or manually type the website you want to access in the browser address/URL bar.

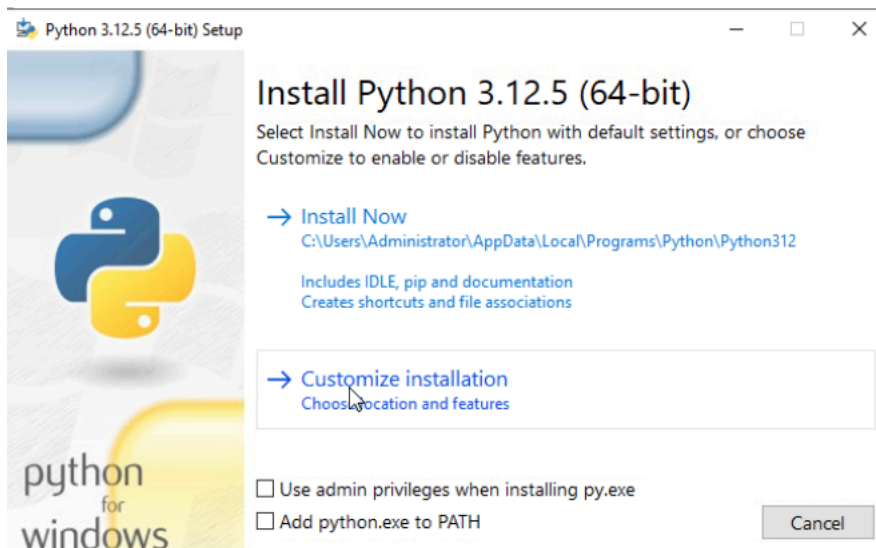
Lab Environment Configuration and Dependencies

Install the Latest Version of Python

Verify that Python 3.x is installed on your system. If Python is not already installed on your system, please download it by typing the following link into your browser: <https://www.python.org/downloads/>.

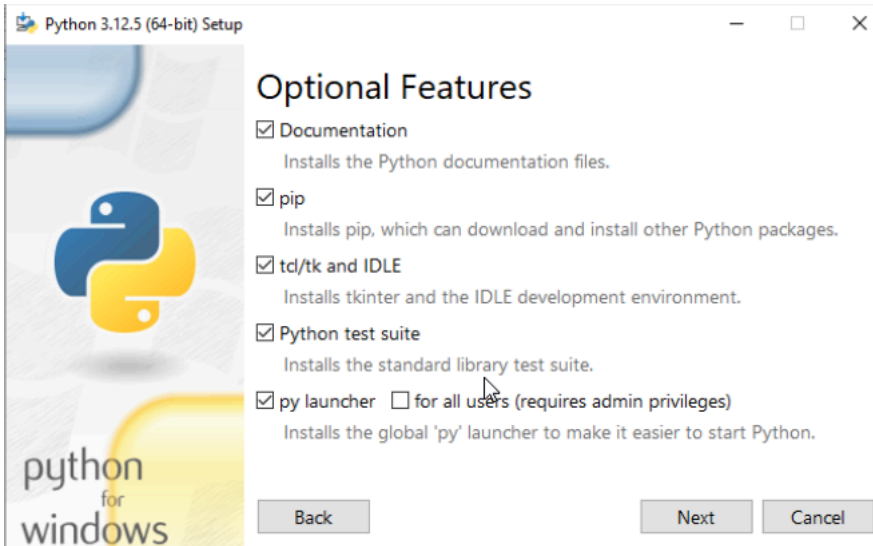


1. While Installing Python for Windows, select **Customize installation**.



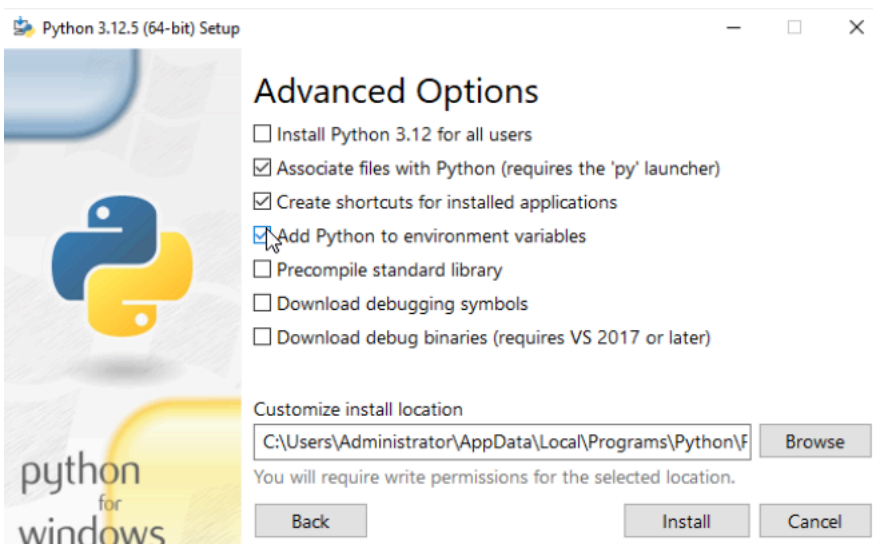
2. On the **Optional Features** window, select the following options and click **Next**.

- Documentation
- pip
- tcl/tk and IDLE
- Python test suite
- py launcher



3. On the **Advanced Options** window, select the following options and click **Install**.

- Associate files with Python (requires the 'py' launcher)
- Create shortcuts for installed applications
- Add Python to environment variables



Install Scapy

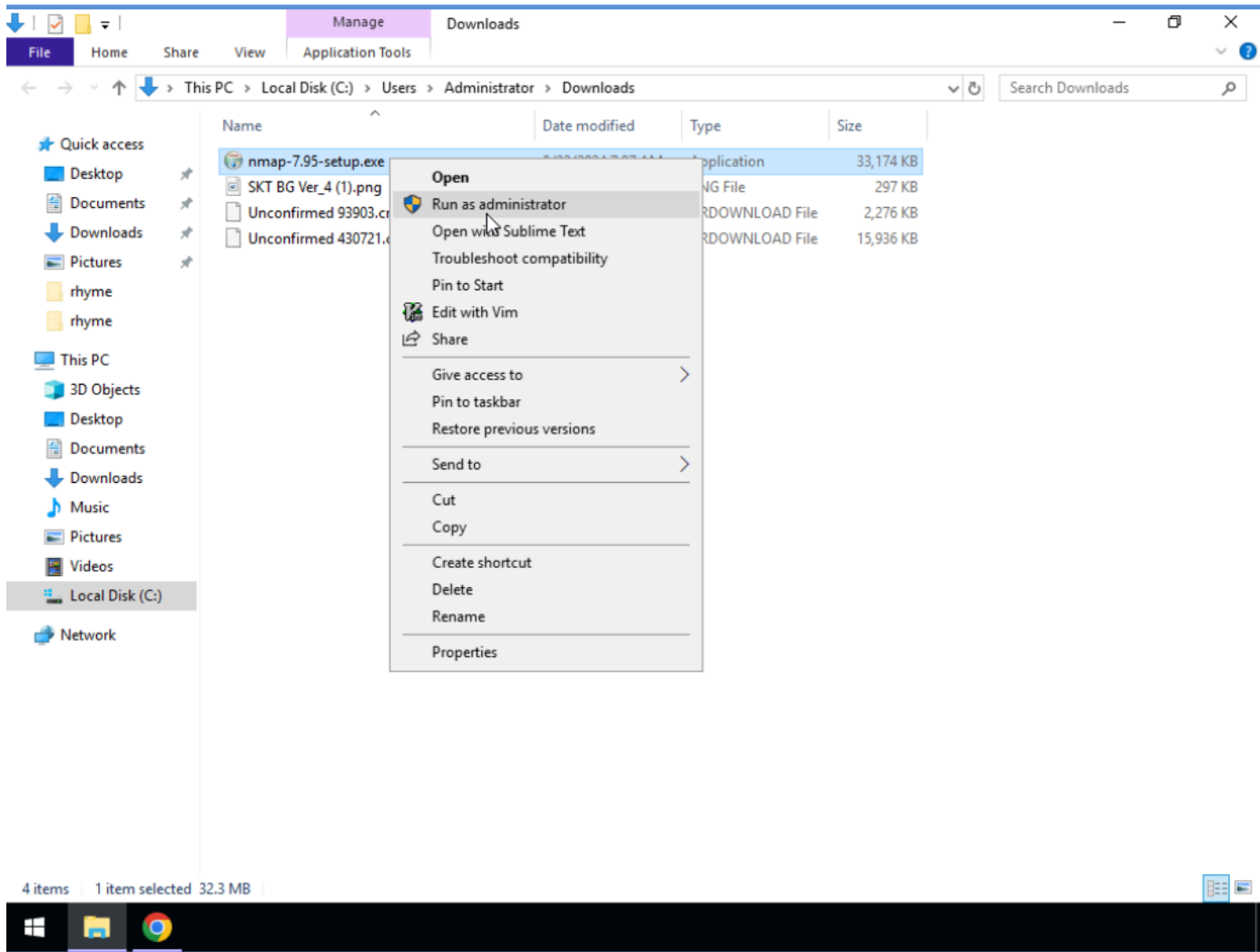
You'll need the Scapy library to use the **Sorst Scan Detector** Python script. Scapy is a powerful Python library used for network packet manipulation and sending and receiving network packets.

- To install Scapy, open a command prompt using the pip command. Type the following command:

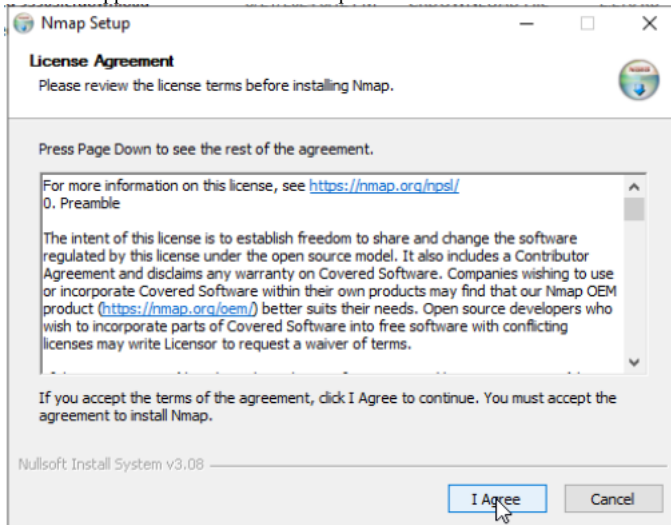
```
pip install scapy
```

Install nmap for Port Scanning

1. Download **nmap** from <https://nmap.org/download>.
2. Verify that you have administrator privileges for the computer.
3. Navigate the folder where you downloaded nmap.
4. Right-click the file name and install the application using administrative privileges.



5. Install the application with all default options selected.



Download the Port Scan Attacker and Detector

You also need to install the Port Scan Attacker and Detetdor Python files. You will download these files to the lab Workspace. The steps are as follows:

1. Open a web browser and downloadd the Port Scan Detect file by typing the following link into your browser: <https://bit.ly/4ed1Bw3>. A file name **port_scan_detect.py** should be downloaded to your windows lab environment.
2. Similary download the Port Scan Attack file by typing the following link into your browser: <https://bit.ly/3TfxhIS>. A file name **port_scan_attack.py** should be downloaded to your windows lab environment.
3. After successfully downloading the files, make sure the files are available on your desktop in the workstation environment during your session.

Port Scanning and Detection

Scenario: After mitigating a Ping of Death attack, you notice unusual network traffic patterns. Suspecting that an attacker might be scanning your network for open ports, you decide to perform a port scan to understand how an attacker might exploit your network. After performing this process, you will verify if your intrusion detection system can detect such scanning activities.

What is Port Scanning?

Port scanning is a technique used by attackers to identify open ports and services running on a target machine. By scanning a range of ports, an attacker can determine which services are available and potentially vulnerable to exploitation. While port scanning is often used for legitimate network management, port scanning is also a common precursor to more serious attacks.

Perform a port scan:

1. In the search bar type **Command Prompt**. Right-click on it and select **Run as Administrator**.
2. In the command prompt window, navigate to the folder where `port_scan_attack.py` is located.

```
cd desktop
```

3. Run the the port scan script by typing the following command in the command prompt window:

```
python port_scan_attack.py
```

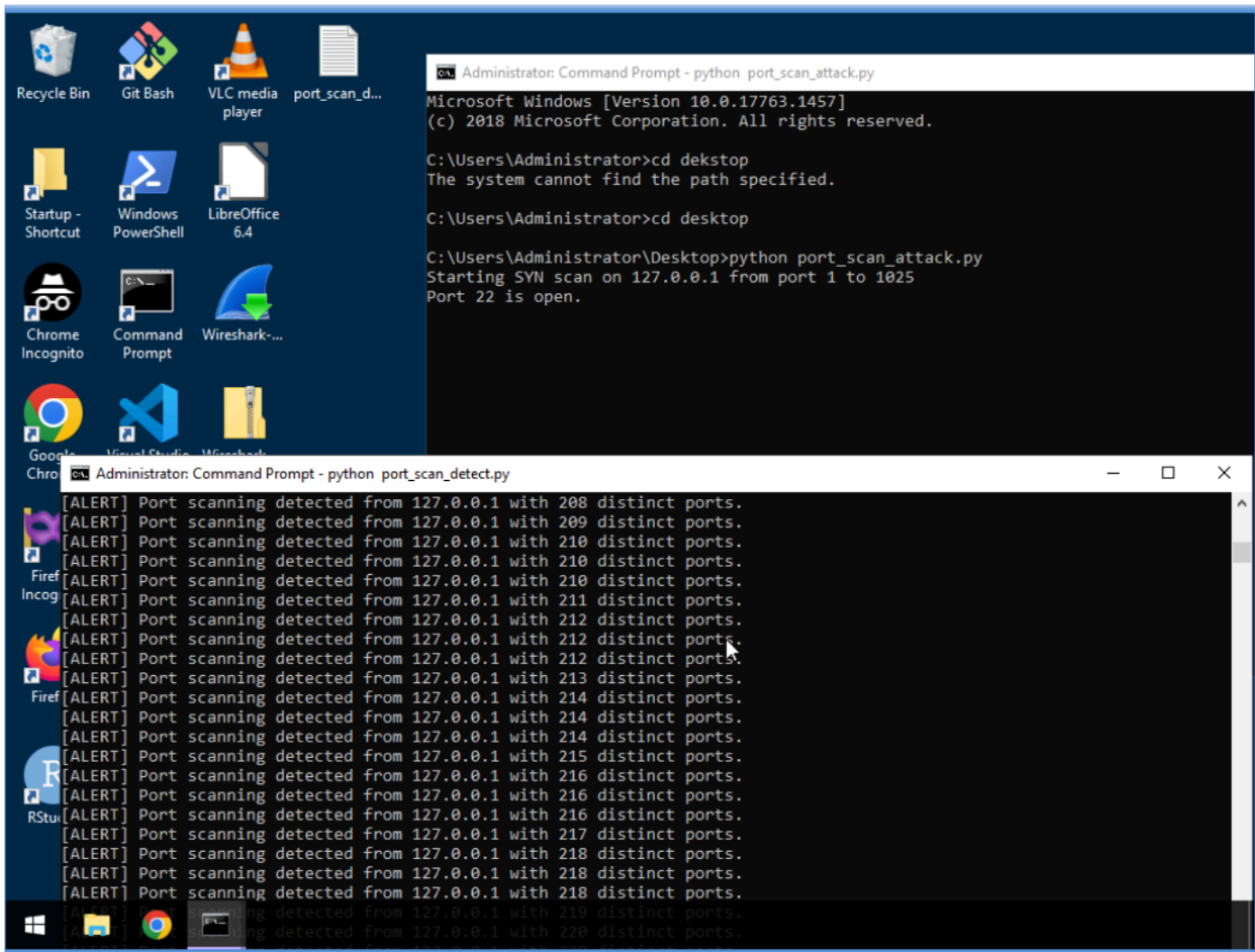
4. Open a second command prompt window.
5. In the command prompt, navigate to the folder where `port_scan_detect.py` is located.

```
cd desktop
```

5. Run the port scan script typing he following command in the command prompt window:

```
python port_scan_detect.py
```

6. Analyze the detection ouput. Review the detection output for any alerts generated by the script.



Exercise

Scenario: In this exercise, you'll combine the power of Nmap, a widely used network scanning tool, with a custom Python script (`port_scan_detect.py`) to perform a local system scan and detect potential port scanning activities.

You will begin by launching **Nmap** with administrative privileges, then run the Python script to monitor the system. With **Nmap** running, you'll initiate a scan on the loopback IP address (127.0.0.1) and analyze the results.

Open Nmap

1. Select **Start** and locate the Nmap icon or search for the Nmap icon on the desktop.
2. Right-click the Nmap icon and select **"Run as administrator"** to open Nmap with administrative privileges.
3. In the search bar type **Command Prompt**. Right-click on it and select **Run as Administrator**.
4. In the command prompt window, navigate to the folder where `port_scan_detect.py` is located using the following command:

```
cd desktop
```

5. Run the port scan script by typing the following command:

```
python port_scan_detect.py
```

Start the Nmap Scan

1. In the Nmap interface (Zenmap), type the loopback IP address `127.0.0.1` in the **Target** field.
2. From the available options, select the type of scan you want to perform such as **Intense scan** or **Quick scan**. Then select the “**Scan**” button to begin scanning the local system.

Review the Results

- Review the detection output for any alerts generated by the script.

Conclusion:

This lab demonstrates how to execute and detect network attacks on a Windows machine. By analyzing the alerts generated during the Ping of Death and port scanning activities, you gain insight into the importance of intrusion detection systems in maintaining network security. Through realistic scenarios, you've observed how these attacks can disrupt network operations and the critical role of detecting and mitigating them. Similarly, industry-standard Intrusion Detection Systems (IDS) operate using signature-based techniques to identify and respond to various types of attacks. These systems maintain signatures for a wide range of known threats, allowing them to recognize and mitigate malicious activities in real time, thereby safeguarding network environments from a multitude of security threats.

Author(s)

Dr. Manish Kumar



Skills Network